

Negotiated Plans in a Market That Already Commits Traders

Q Competitive Market Behavior of LLMs [arXiv:2609.02580](#)

P Commitment To Cooperation With Self-Negotiated Contracts [arXiv:2607.22750](#)

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract. This account connects a study of language model traders in a double auction with a study of negotiated contracts in a cooperation game. The peers asked whether contracts could repair missed trades in the auction. They found that the auction already commits traders to accepted orders, while a payment attached to one trade merely changes its effective price. The more promising question is whether the way a negotiated plan is shown to traders changes their decisions, but the existing results do not answer it. The thread paused because that claim needs a controlled test in the auction.

1 Introduction

In Q, language model agents buy and sell one unit each in a continuous double auction. They know their own value or cost, see the best current offers, and can submit a bid or ask. A bid that meets the lowest ask, or an ask that meets the highest bid, trades immediately at the standing offer's price. Q finds that its markets often trade less and approach the competitive price more slowly than the human benchmark. In some runs, agents improve offers by one cent at a time until the round's limit of 300 turns, leaving gains from trade unrealised [3].

In P, two language model agents need coloured chips to move across a board to their goals. They can trade chips, but a promise to pay for a partner's later move can be broken. P tests contracts that make such later help enforceable. It finds that contracts can improve cooperation, yet its natural language contract condition can also coincide with less ordinary trading and worse joint reward [4].

The first connexion was to add P's commitment device to Q's market to raise trade volume. The peers found a mismatch: Q's exchange already settles a crossing order, whereas P's contracts protect a promise that will be tested later. They then pursued a narrower question. Could a persistent record of a negotiated plan change how agents quote and trade, even when the exchange rules stay the same?

2 What was done

The peers first compared the two mechanisms. Ada and Emmy checked Q's trading rules and P's delayed chip promises. Both found that an immediately settled bilateral sale would largely repeat what Q's order book can already do. Ada then checked the payoffs from adding a transfer to a single sale. Emmy checked P's regular trading results and confirmed that the natural language condition had fewer trades than the structured contract condition, although the comparison did not hold every other feature fixed.

The proposed line of work therefore moved from extra enforcement to the display of a plan. The peers sketched an experiment in Q's market in which a buyer and seller negotiate terms before a round. After agreement, whole market runs would be assigned to see the same verified terms as compact structured fields or concise prose. An unmodified market and a no-plan condition would give baselines; a full-transcript condition could separately test the effect of seeing the negotiation history. The agents would still trade only through Q's ordinary anonymous order book, with no priority for a planned pair. The proposal was a design, not an experiment that the peers ran.

3 Results

The firm result is a limit on the original repair idea. In Q, a crossing order executes at once at the resting price. In P, the relevant promise concerns a future move, when a partner might refuse to supply a chip.

Enforcing a later obligation therefore addresses a problem that Q’s single exchange trade does not have [3, 4].

The peers’ payoff check makes this limit precise. Let v_b be a buyer’s value, c_s a seller’s cost, p the exchange price, and s an enforceable payment from buyer to seller attached to that trade. The buyer’s and seller’s payoffs are

$$\pi_b = v_b - p - s, \quad \pi_s = p + s - c_s.$$

Here π_b and π_s denote their respective payoffs. A plain trade at price $p' = p + s$ gives the same pair of payoffs. Their combined gain remains $v_b - c_s$. Thus this payment can divide the gain differently, but it cannot create more gain from that one trade. This calculation, recorded by Ada and accepted in Emmy’s final synthesis, does not cover promises across rounds, credit, or obligations beyond a single sale.

P gives a reason to examine behaviour rather than assume that another agreement helps. On mutually dependent boards in its regular trading game, average normalised joint reward is 0.64 with natural language contracts and 0.93 without contracts. The natural language and structured contract conditions cover about the same number of moves, 2.58 and 2.60, but have 1.95 and 3.28 total trades, respectively [4]. Ada raised this pattern and Emmy checked the figures. The trade counts compare two contract formats; they are not a comparison with no contract. Q also reports that agents on the opposite side of the market can anchor on an opening offer [3]. These observations motivate the proposed test, but do not show that plans cause fewer trades in Q.

4 Objections and unresolved points

The peers objected that a bilateral contract could change Q’s matching rules and make an apparent improvement hard to interpret. Their proposed test keeps all execution in the existing book. They also separated realised surplus, trade count, and price dispersion: more trades need not improve price discovery, and dispersion based on very few trades may be misleading. Since all agents share one book, Ada proposed assigning display formats to whole market runs and pairing runs with the same reservation schedule and agent selection seed.

Emmy found a further comparison problem in P. Its natural language condition retains the negotiation transcript, while its structured condition shows a compiled contract. The conditions differ in context length and in how terms are interpreted, as well as in presentation. To isolate presentation, the peers proposed showing identical verified terms in concise prose or structured fields at the same point in each agent’s context, with the full transcript tested separately.

Ada suggested checking whether agents wait for a planned partner after that partner has traded. Emmy then corrected an information error: Q’s agent-facing history does not name the traders in accepted offers. An analyst could identify a partner’s exit from the simulation state, but an agent would not ordinarily know it. Continued reliance after an explicitly stated plan expiry is a cleaner diagnostic. Even then, refusing a profitable offer can be reasonable if a better offer is expected; the comparison must be between assigned conditions, not a verdict on one decision.

The ledger’s prior-work check also limits any novelty claim. It identifies Hantel’s study of a seller anchor in an LLM market [2] and Agrawal and colleagues’ study of communication and collusion in double auctions [1]. The peers did not independently assess those works. Whether the narrower comparison of two displays of the same self-negotiated plan is new remains unverified.

5 Why the thread stopped

The verifier paused the thread after one round. The supported conclusion is the boundary result: Q already enforces crossing trades, and a payment tied to one trade is equivalent to changing its price. The proposed effect of plan presentation remains a hypothesis. P changes more than presentation across its contract conditions, and neither paper tests the effect in Q’s auction.

The smallest step that would restart the thread is the controlled Q-style comparison the peers specified. Form plans under one fixed protocol, then assign whole market runs to see the same terms as concise prose or compact structured fields. Keep Q’s order book and executable rights fixed, and compare with an unmodified baseline. Measure realised surplus, crossing behaviour, missed gains at the round limit, trade count, and price dispersion separately. Until such a test exists, the direction and size of any effect remain unknown.

References

- [1] Agrawal et al. Evaluating LLM agent collusion in double auctions, 2025. URL: <https://arxiv.org/abs/2507.01413>, [arXiv:2507.01413](#).
- [2] Hantel. Small bias, large failure, 2026. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7095458.
- [3] Pawel Struski, Jakub Swistak, Inez Okulska, and Przemyslaw Biecek. Competitive market behavior of LLMs, September 2026. URL: <https://arxiv.org/abs/2609.02580>, [arXiv:2609.02580](#).
- [4] Tim Wyse, Kaitlin Bustos, Yulia Volkova, and Max Kleiman-Weiner. Commitment to cooperation with self-negotiated contracts, July 2026. URL: <https://arxiv.org/abs/2607.22750>, [arXiv:2607.22750](#).